

Data Structures and Algorithms – Spring 2026

(BS-IT-F24 Morning & Afternoon)

Course code	Course (CC-213)	Lab (CC-213-L)
Credit hours	Course (3)	Lab (1)
Prerequisite	Object-Oriented Programming (CC-211)	
Course Instructor	Muhammad Ahmad Ghazali Email: ahmad.ghazali@pucit.edu.pk	

Course Objectives

- To introduce data structures as basic building blocks of large programs.
- To learn the commonly used data structures.
- To be able to implement the commonly used data structures in C++.
- To introduce the notion of time and space complexity.
- To develop the skills to analyze time and space requirements for a data structure and associated algorithms.
- To prepare the students to pick the right data structure for a given problem.

Textbook

- E. Horowitz, S. Sahni, and D. Mehta, *Fundamentals of Data Structures in C++*, 2nd Edition Silicon Press/Universities Press, 2007.

Reference Books

- A. Drozdek, *Data Structures and Algorithms in C++*, 4th Edition, Cengage Learning, 2013.
- L. Nyhoff, *ADTs, Data Structures, and Problem Solving with C++*, 2nd Edition, Pearson, 2005.
- M.A. Weiss, *Data Structures and Algorithm Analysis in C++*, 4th Edition, Pearson, 2014.
- N. Dale, *C++ Plus Data Structures*, 3rd Edition, Jones and Bartlett, 2003.

Grading Scheme (Tentative)

<u>Course</u>		<u>Lab</u>	
Quizzes	14%	Lab Tests	14%
Assignments	08%	Assignments	08%
Homeworks	03%	Lab Tasks	28%
Midterm Exam	35%	Midterm Exam	20%
Final Exam	40%	Final Exam	30%

Passing Criteria

- Minimum requirement to pass this Course and Lab is to get overall 50% marks in Course as well as in Lab.

Tentative Course Outline and Lecture Plan

Topics	No. of Lectures
Introduction	1
List ADT: Unsorted and Sorted	2
Performance analysis and measurement, Time complexity, Space complexity	2
Searching – Linear search, Binary search, and their analysis	1
Sorting – Bubble sort, Selection sort, Insertion sort, and their analysis	1
Stacks using arrays, Evaluation of Expressions	2
Queues using arrays	1
Recursion – Examples, Analysis of recursive algorithms	2
Recursive Sorting algorithms – Merge sort, Quick sort, and their analysis	2
Linked lists – Motivation and Implementation of Singly-linked lists	2
Midterm Exam	
Stacks and Queues using Linked lists, Recursion on Linked lists	2
Circular and Doubly-Linked lists	1
Trees – Binary Trees and Traversals	2
Binary Search Trees – Search, Insert, Delete	2
Heaps and Priority Queues, Heap sort and its analysis	2
Height Balanced Trees (AVL Trees or 2-3 Trees)	1
Hashing	1
Graphs – Representation, Traversals (DFS and BFS)	1
Final Exam	

Important Notes

- Academic integrity is expected of all students. Plagiarism/cheating or usage of AI tools in any assessment will result in at least an "F" grade in the course, and possibly more severe penalties.
- You bear all the responsibility for protecting your Assignments/Labs etc. from plagiarism. If anyone else submits your Assignment/Lab or uses your code in his/her Assignment/Lab, you will be considered equally responsible and will be punished equally.
- The instructor reserves the right to modify the grading scheme/marks division and course outline during the semester.
- Knowledge of C++ and Object-Oriented paradigm is assumed for this course. All code written in Quizzes, Assignments, Homeworks, Labs, and Exams MUST be in C++. Code must be intelligently commented. Uncommented code may not be given any credit.
- You are required to use Microsoft Visual Studio for writing your code.
- There is **NO makeup** for a missed Quiz, Lab test, Assignment, Lab, or Homework.